

WORK FLOW

Input Image



Object Detection (YOLO) + Scene Classification (ResNet18)
(performed in parallel)



Extract Objects & Scene Labels



Map to Predefined Categories



Compute Weighted Scores



Select Highest Score



Final Image Category (Hospital/Classroom/Roadside/etc.)

ALGORITHM

Input: An input image **I**

Output: Final category label **C** (e.g., Hospital, Classroom, Roadside)

Step 1: Object Detection

1. Load the pretrained **YOLO** object detection model.
2. Pass the input image **I** to the YOLO model.
3. Detect all objects present in the image.
4. Store the detected object labels along with their confidence scores.

Step 2: Scene/Background Classification

1. Load the pretrained **ResNet18 (Places365)** scene classification model.
2. Preprocess the input image (resize, normalize, and convert to tensor).
3. Pass the preprocessed image to the model.
4. Obtain the scene prediction probabilities.
5. Select the top predicted scene labels with their confidence scores.

Step 3: Define Category Mapping

1. Define the target image categories (e.g., Hospital, Classroom, Roadside).
2. Assign the relevant object labels to each category.
3. Assign the relevant scene labels to each category.

Step 4: Score Computation

1. Initialize the score of every category to zero.
2. Compare the detected objects with the predefined object labels of each category and update the scores.
3. Compare the predicted scene labels with the predefined scene labels of each category and update the scores.
4. Combine the object score and scene score using predefined weights.

5. Step 5: Decision Rule

1. Select the category with the highest combined score.
2. If no category receives a valid score, classify the image as **"Uncategorized."**
3. Return the final predicted category.